

Formula

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, \nu} \cdot$$

$$\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

with ν from the Welch-Satterthwaite formula.

Example

Streaming data: difference = 4, SE = $\sqrt{4.1^2/14 + 6.3^2/10} \approx$

$$3.33, \quad \nu \approx 16.3, \quad t^* \approx 2.12 \Rightarrow 4 \pm 2.12 \cdot 3.33 = (-2.1, 10.1).$$

Interpretation

Interval includes 0 $\Rightarrow$ cannot conclude a difference at $\alpha =$

$$0.05.$$